

HACKATHON 2026 | INNOVANDO POR UN FUTURO SOSTENIBLE

CarbonSmart Buildings

Smart Platform for Carbon Footprint Reduction
and Tokenization on LACNet

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Sustainability &
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37%

of global energy-related CO₂ emissions originate from residential, commercial, and institutional buildings.

High electricity consumption in lighting, HVAC systems, and elevators creates an unsustainable footprint. Current facilities completely lack the tools to measure, verify, and reward effective reduction.



The Market Failure of Building Emissions



Information Asymmetry

A complete lack of reliable, continuous, and verifiable data regarding actual energy consumption and building emissions.



Negative Externalities

Excessive fossil-based electricity consumption creates massive societal costs that are entirely unreflected in the administrative energy price.

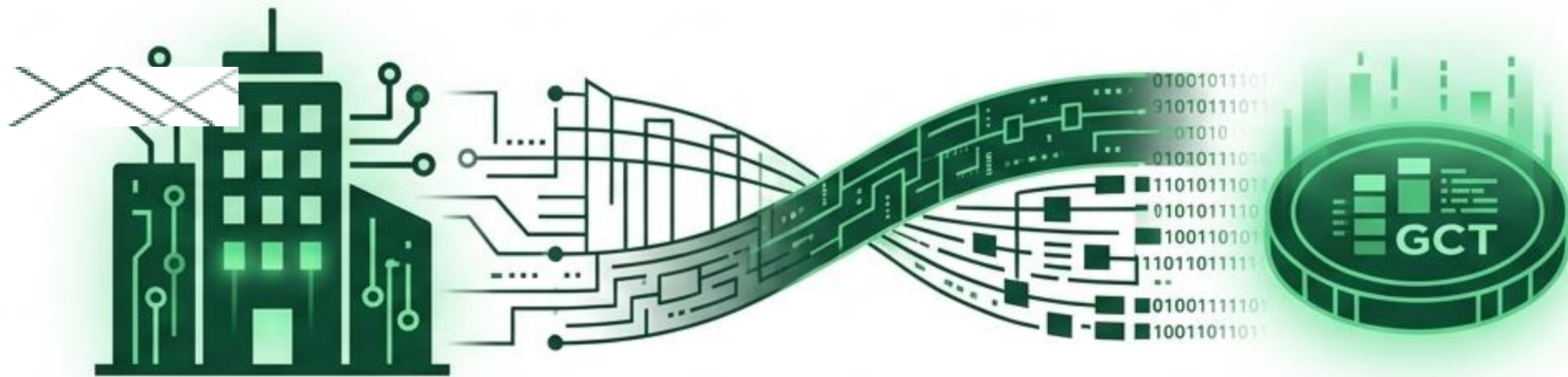


High Monitoring Costs

Traditional energy audits demand specialized equipment, manual technical inspections, and prohibitive administrative friction.

CarbonSmart: A Digital Pigouvian Subsidy

By utilizing IoT, AI, and Blockchain, CarbonSmart continuously monitors energy consumption and automatically rewards verifiable footprint reductions.



When a building demonstrably reduces emissions against a certified baseline, Smart Contracts activate.

The protocol automatically releases Green Carbon Tokens (GCT).

This creates a direct, automated financial incentive that financially internalizes positive ecological behavior.

The CarbonSmart Engine



01. Real-Time Source Monitoring

IoT Stack (Smart energy meters, occupancy, temp/humidity, and air quality sensors) captures consumption data instantly, eliminating manual audits.

02. Cryptographic & AI Validation

Oracles feed data to AI models that detect consumption patterns and calculate exact carbon footprint reductions against baselines.

03. On-Chain Minting

The LACNet-compatible Smart Contract automatically mints Green Carbon Tokens (GCT) directly to the building administrator's wallet.

The Trust Engine Architecture



Layer 1 (The Eyes): IoT Integration Layer

Smart sensors deployed throughout the building gather raw, localized environmental data.



Layer 2 (The Brain): AI & Oracle Layer

Artificial Intelligence processes the data for anomaly detection and efficiency recommendations. Oracles securely bridge this off-chain data to the network.



Layer 3 (The Ledger): LACNet Settlement Layer

The immutable blockchain layer where CarbonSmartToken.sol and EnergyRewardManager.sol execute conditional distribution based purely on cryptographic proof.

Protocol Architecture & Immutability



LACNet Infrastructure

Native deployment on the enterprise-grade LACNet regional network, ensuring high throughput and low-cost execution.



Solidity Smart Contracts

Self-executing contracts utilizing BaseRelayRecipient for meta-transactions, removing gas-fee barriers for end-users.



Token Taxonomy Framework (TTF)

Ensuring the Green Carbon Token is interoperable, standardized, and ready for institutional adoption.



IPFS Decentralized Storage

Energy audits and environmental evidence are stored via IPFS, linked by on-chain hashes to guarantee absolute data integrity.

The Paradigm Shift in Environmental Auditing

	Traditional Methods	CarbonSmart Protocol
Data Frequency	Periodic, manual, snapshot-based.	Continuous, real-time, 24/7 monitoring.
Trust Mechanism	Centralized third-party inspectors (prone to friction/fraud).	Trustless cryptographic validation & AI Oracles.
Cost Structure	High, linear marginal costs for every new building audited.	Marginal cost tends toward zero after initial deployment.
Incentive Alignment	Punitive (fines for non-compliance).	Pigouvian (automated financial rewards for positive behavior).

Adoption Without Technological Barriers

Web3 Authentication

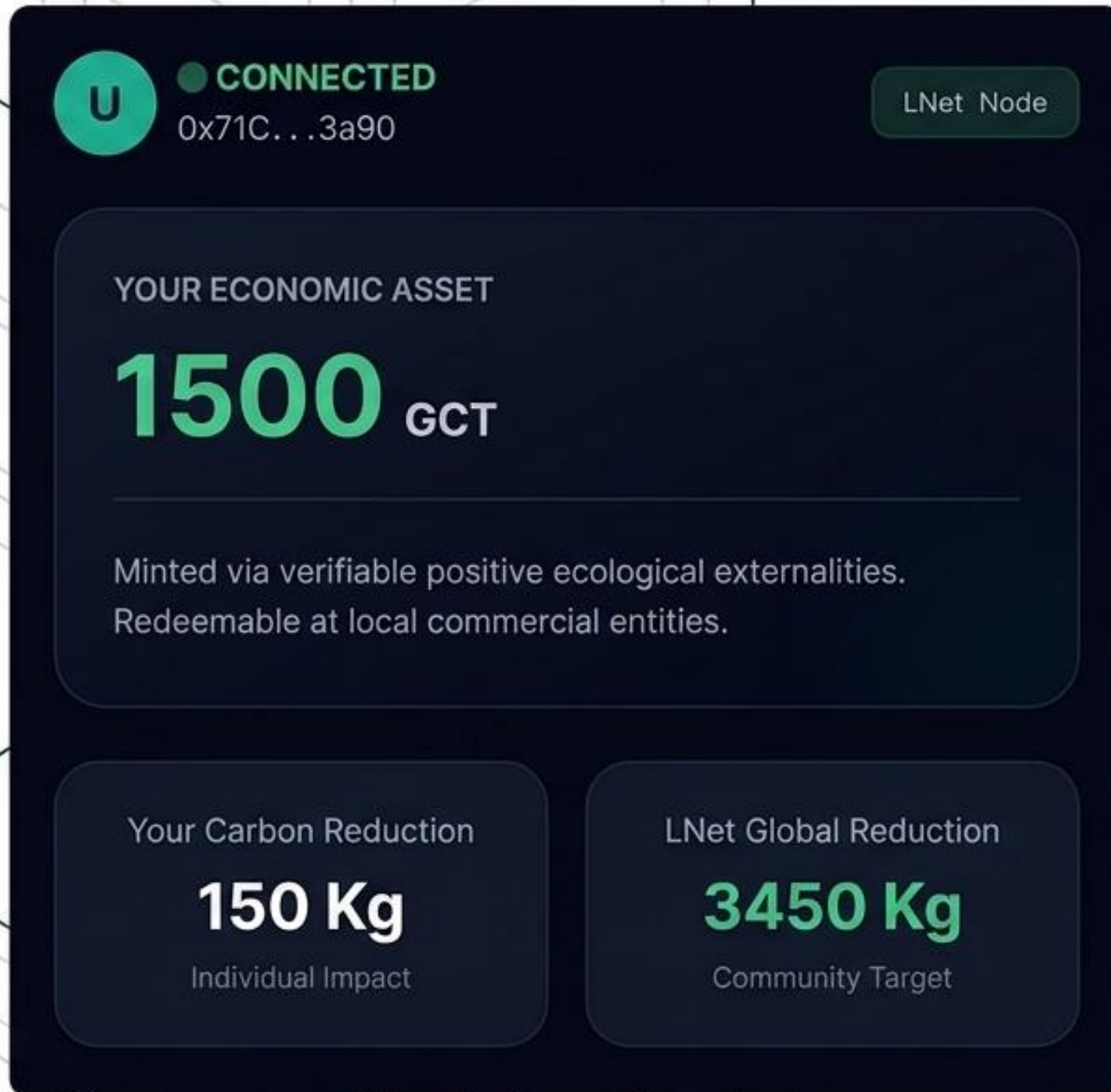
Secure access without centralized passwords.

Real-Time Metrics

Visualizing exact CO₂ reductions and energy performance.

Token Balance

Live view of accumulated Green Carbon Tokens.



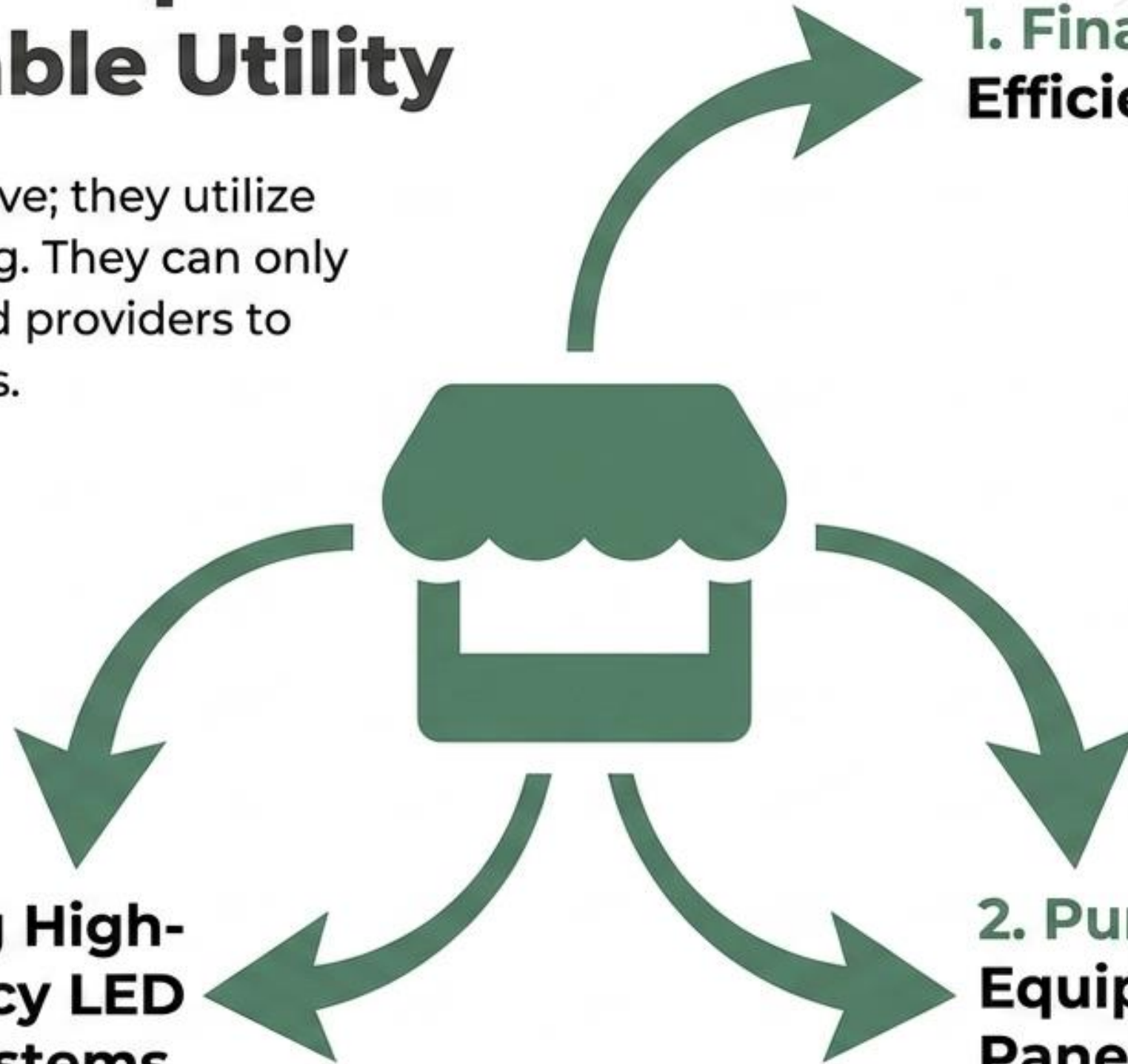
The EcoMarketplace: Programmable Utility

Tokens are not speculative; they utilize Programmable Spending. They can only be spent with authorized providers to further reduce emissions.

3. Installing High-Efficiency LED Lighting systems.

1. Financing Energy Efficiency Upgrades.

2. Purchasing Sustainable Equipment (e.g., Solar Panels, advanced HVAC).



Tokenomics: The Triple Value Model



Environmental Value

Strictly linked to **mathematically** verifiable CO₂ emission reductions (no reduction, no token).



Utility Value

Direct purchasing power via the EcoMarketplace for sustainable reinvestment.

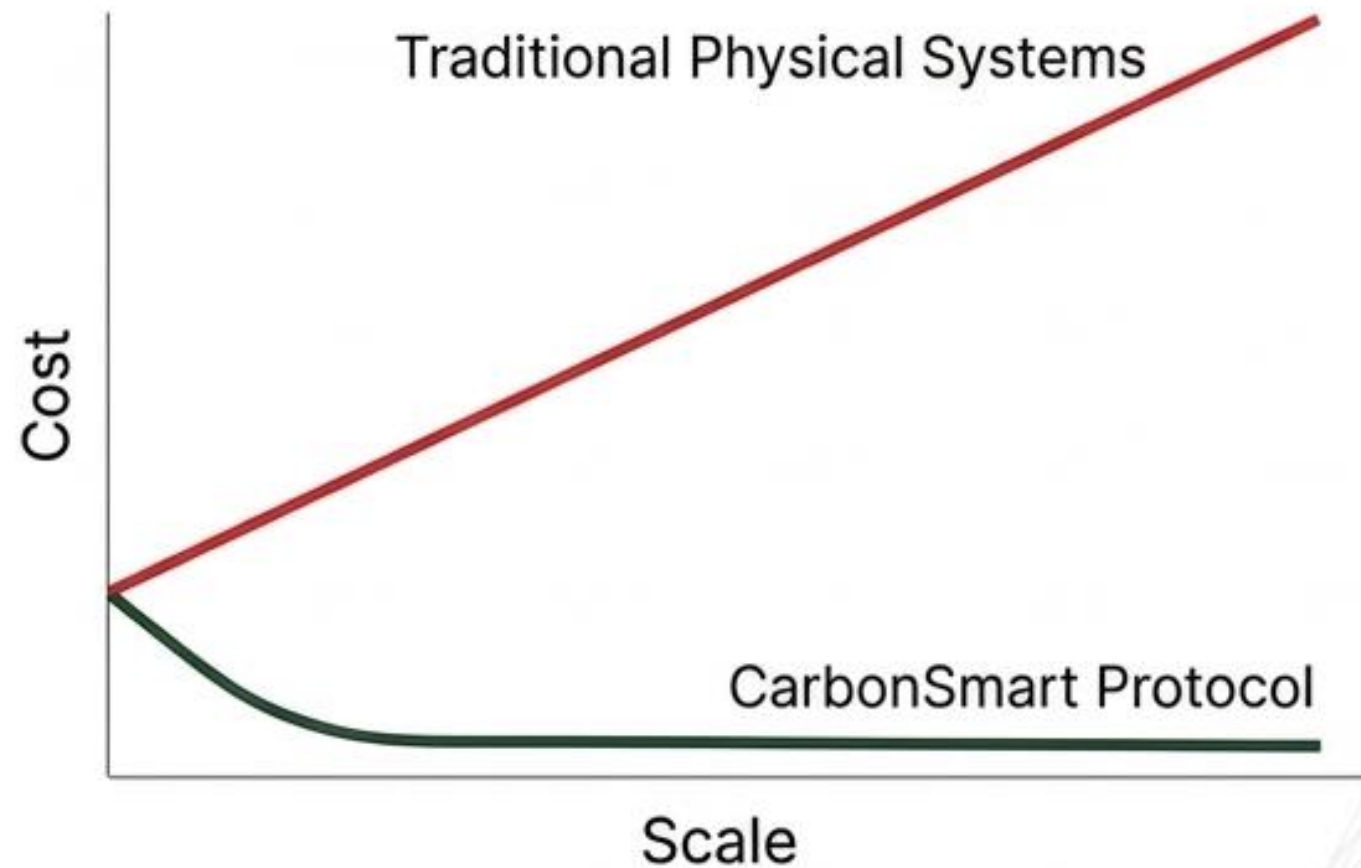


Network Value

Generating a transparent, interoperable, and fully traceable ledger of regional environmental data.

Anti-Fraud Mechanism: Conditional distribution ensures tokens are only minted when IoT and AI models cryptographically verify the baseline reduction.

Economic Principles: Scale and Signaling



Marginal Cost Tending to Zero ($MC \rightarrow 0$):

While traditional physical systems get more expensive linearly as they expand, CarbonSmart scales instantly on LACNet's distributed nodes with near-zero marginal cost.

Signaling Theory (Spence, 1973)

The public, immutable blockchain record acts as a credible, auditable "green signal." It allows businesses to publicly verify their ecological compliance to an ecological matations to investors and regulators without requiring costly third-party audits.

Alignment with Global Hackathon Objectives



Global Specification Automation (GSA)

Delivering TTF-compatible Smart Contracts with adaptable, automated incentive rules powered by IoT validation.



Multi-Nation Token Management (MNT)

Creating interoperable Green Carbon Tokens designed for integration with regional blockchain networks across Latin America.



International Compliance & Reporting (ICR)

Generating automated energy performance reports backed by IPFS evidence, guaranteeing transparency for international regulators.

Projection & Regional Deployment Plan



Phase 1: Hackathon (Q3 2026)

Smart Contract deployment on LACNet testnet, Green Carbon Token minting logic, and functional IoT dashboard.

Phase 2: Pilot (Months 1-6)

Deployment in a selected university/corporate building in Peru. Full API integration and automated emission certification.

Phase 3: National Scale (Months 6-18)

Advanced AI predictive deployment. Scaling to 100+ smart buildings and integrating with national energy efficiency systems.

Phase 4: LatAm Expansion (Years 2-3)

Regional deployment across LACNet, establishing partnerships with construction firms, and consolidating as the Latin American standard for building tokenization.

Open-Source Infrastructure as a Regional Public Good

*Built on the principles of non-rivalry and non-excludability
to enable cumulative, sustainable innovation across
municipalities and smart cities in Latin America.*